NAMD-AUTOMATOR

Version 1.0

USER MANUAL

Dr Rohan J Meshram

Mr. Manojit Mazumder

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1. Introduction

Welcome to **NAMD-Automator**, a user-friendly graphical interface designed to simplify the process of Molecular Dynamics (MD) Simulation, by creating configuration files, and automate the simulation with simple form-based tabs.

The applications of MD simulations are included but not limited to the fields like, Medicinal and Drug Chemistry, Computer aided drug design, Pharmaceutical chemistry, etc. MD simulations offers to model the motions involved in the bio-macromolecules like Proteins, Lipids, Nucleic Acids, etc. Therefore such insights on the movement and interaction of atoms and molecules over time offers unprecedented insights of therapeutic significance. NAMD is one of the widely used software for running these simulations, known for its scalability, speed and efficiency with large molecular systems. However, setting up a typical simulation in NAMD usually requires crafting configuration file for each stage of a Conventional MD Simulation (CMD), such as Minimization, Heating, Equilibration and Production. Following a detailed workflow after creating the files, deciding what should be the content of these files and how to execute the NAMD in command line interface can be challenging for users who are not familiar with scripting or command-line tools.

Keeping this in mind, NAMD-Automator was created to make the process easier and more accessible. It guides users through each step of the simulation such as minimization, heating, equilibration and production through an intuitive tabbed interface. Be it a beginner, or an experienced researcher, this tool helps them set up simulations quickly, avoid common errors in creating the files, syntax error while running the simulation as the process is automated and focus more on the scientific goals rather than technical setup. This manual will walk users through how to use NAMD-Automator effectively, from installation to running a complete simulation, with pictorial representations of each step.

2. System Requirements

The current version of NAMD-Automator is designed to work efficiently in Windows Operating System. The following are major requirements for successful installation of this software

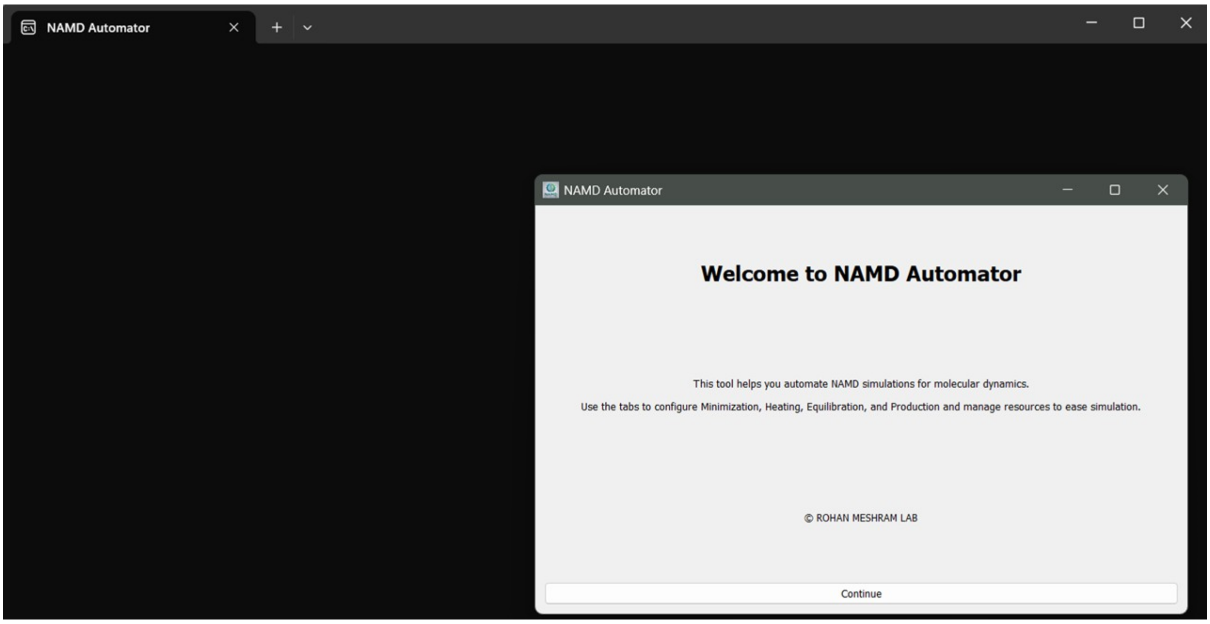
1. Python version 3 and above
2. NAMD (Latest version of Multicore NAMD executables)
   Preinstalled NAMD in C drive. (It is mandatory to have the following path to NAMD application- C:\NAMD\namd2.exe). If user intends to use the GPU enabled version of NAMD, then appropriate CUDA libraries are needed to be installed with nVIDIA graphics card.
3. Windows Operating System 10 and above. We recommend Windows 11 for best user experience.
4. System with at least 4 GB RAM (8 GB recommended), and latest i7 processor is recommended for fast NAMD calculation.

3. Installation

Upon successful downloading the Installer from the Source Forge website. Double-clicking the installer initiates the installation process. Follow the steps in the installation process and the software will be installed accordingly in the system.

4. Usage and Applicability of the tool

After successful installation of the software, a shortcut is made available on the desktop of the system. Double-clicking the shortcut launches the application. When the application is launched it opens with NAMD-Automator Main Window and a command prompt window as follows.



Hitting the continue button will bring up the menu-based form in the main window. The NAMD-Automator software is a tab-based application. Each tab offers a specific utility and features to perform certain tasks related to MD simulations. There are seven tabs in the main window as follows

1. Set Directory
2. Minimization
3. Heating
4. Equilibration
5. Production
6. Manage Resources
7. Help

Obtaining the necessary input files for NAMD-Automator

For conducting the MD simulation in NAMD. The following files are required

1. A solvated system with charges neutralized in the form of PDB-PSF pair. PDB file contains the coordinates of all the atoms in the system; while, the PSF file holds information of connectivity, atom typing, etc. Such PDB-PSF pair can be constructed using various software like VMD. An excellent tutorial from our lab is available on

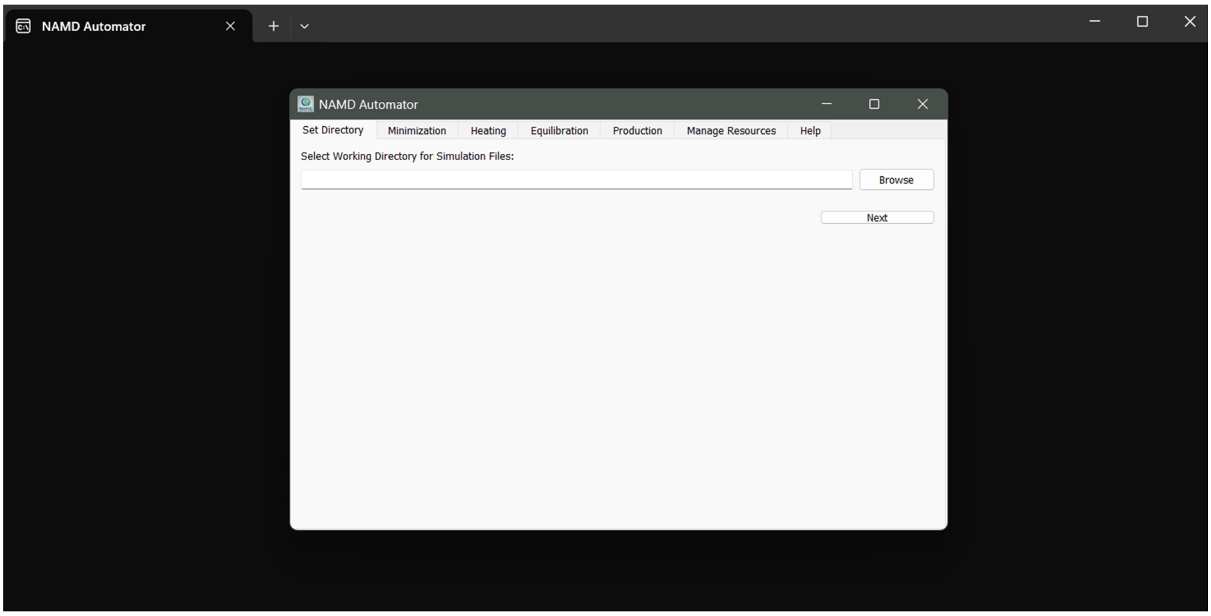
YouTube that guides the user to generate PDB-PSF pair from scratch. The tutorial can be accessed using the url: https://youtu.be/YknrIVrOaAU?si=8ZVlzyfJxeC_HdYo and https://youtu.be/YKNHJmFkLH0?si=OJkzI77PNA7snuHj

2. The user needs to have the parameters of protein, and non-protein components (if any). It is to be noted that the parameters should be compatible to NAMD software. The parameters of the ligands and other non-protein components can be obtained using Swiss-Param server.
3. Additionally, the user is required to fill in the form specifying Periodic Boundary Conditions (PBC), values for setting PME (these values are dependent on PBC), and, coordinates of Cell Origin. We are providing a separate TCL script for efficient calculation of these values along with the software distribution.

Here we assume that, the user has these files (PDB-PSF pair) and the aforementioned values of PBC, PME and Cell Origin for the subsequent procedure.

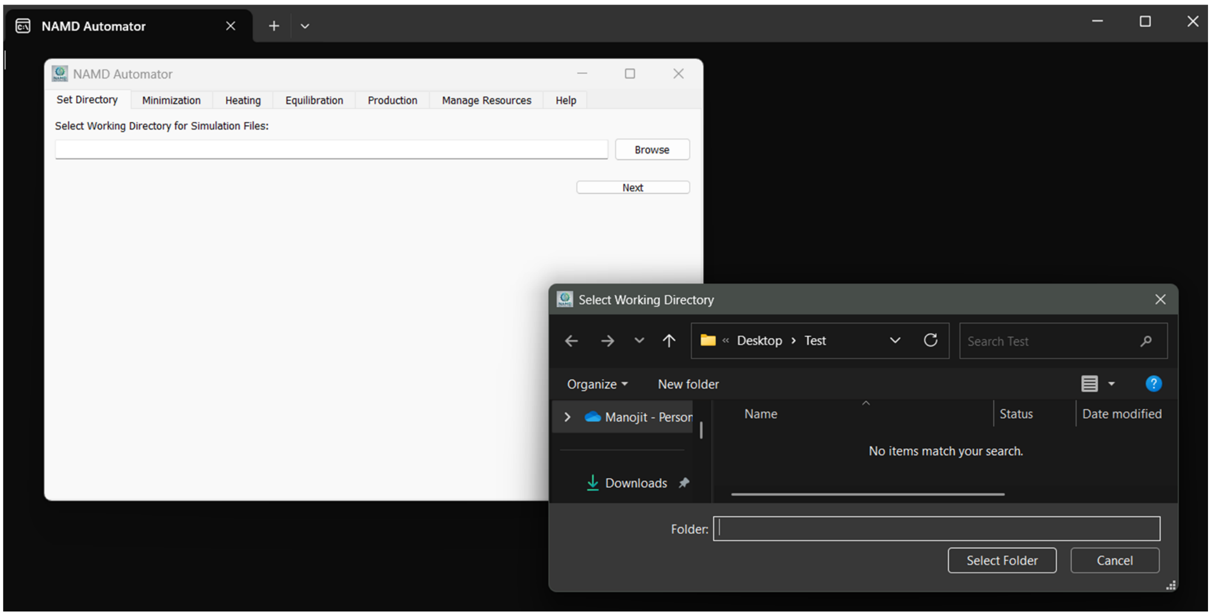
1. Set Directory Tab

For successful usage of this software, the user needs to select the working directory. A working directory is a folder in the system that will hold the required input files and data for the software. By default, the software will look into this folder to locate the input files. The calculated output files will be placed and updated in the same folder. The output files like various configuration files, which will be generated by NAMD-Automator, as well as files created while running the MD simulations by NAMD will be located in this working directory. The "Set Directory Tab" is used to define and set such working directory. The working directory can be set by using the "Browse" button and selecting a desired user defined folder as shown in the figure below.

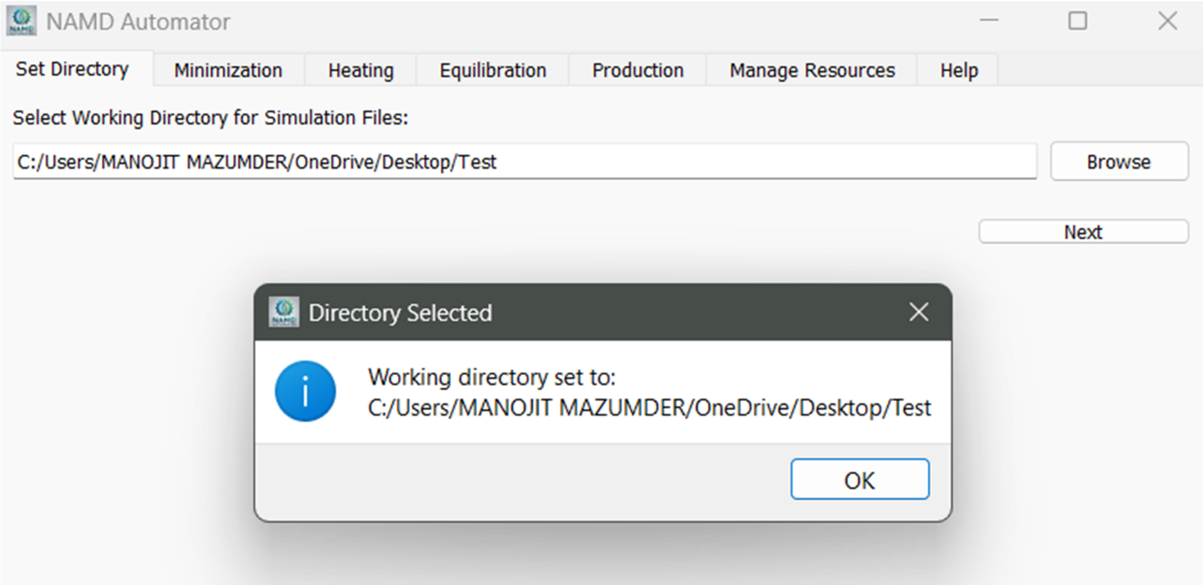


Follow the steps provided below for setting working directory

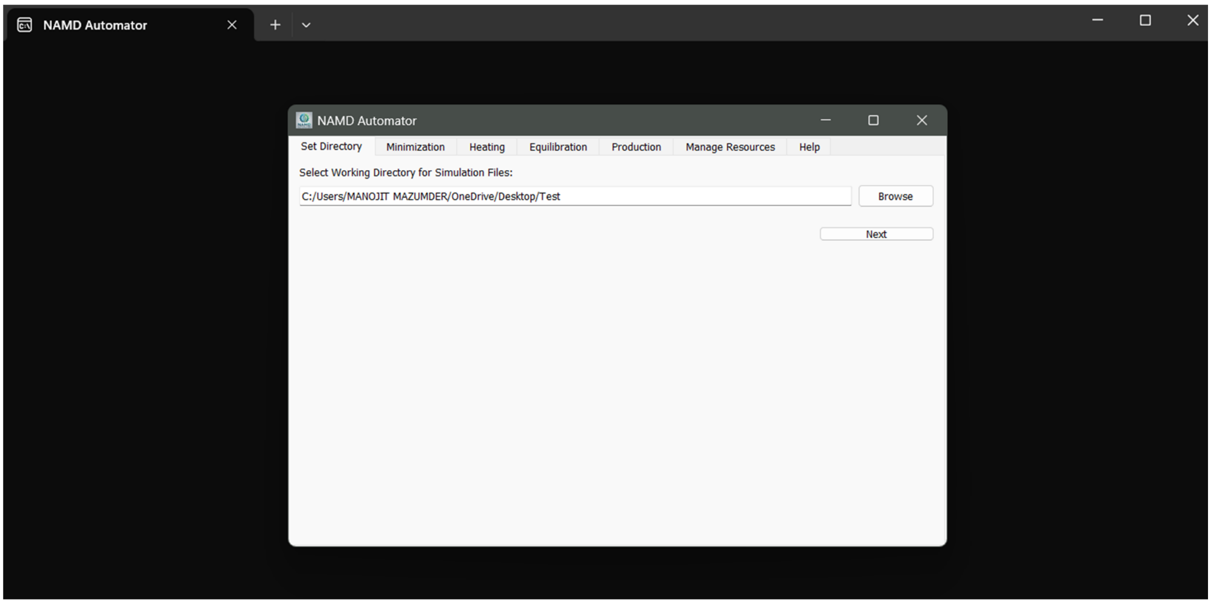
1. Click the "Browse" button in the "Set Directory Tab". A new dialogue box appears as shown in figure below.



2. Choose an appropriate folder by **<u>double clicking</u>** and opening the folder in the software and hit the "Select Folder" button to set the folder as a working directory. Right after selecting the folder to work on, a dialogue box appears saying "Working directory set to: Path_Selected_Working_Directory". Clicking "OK" sets the working directory to the selected one (shown in figure below).



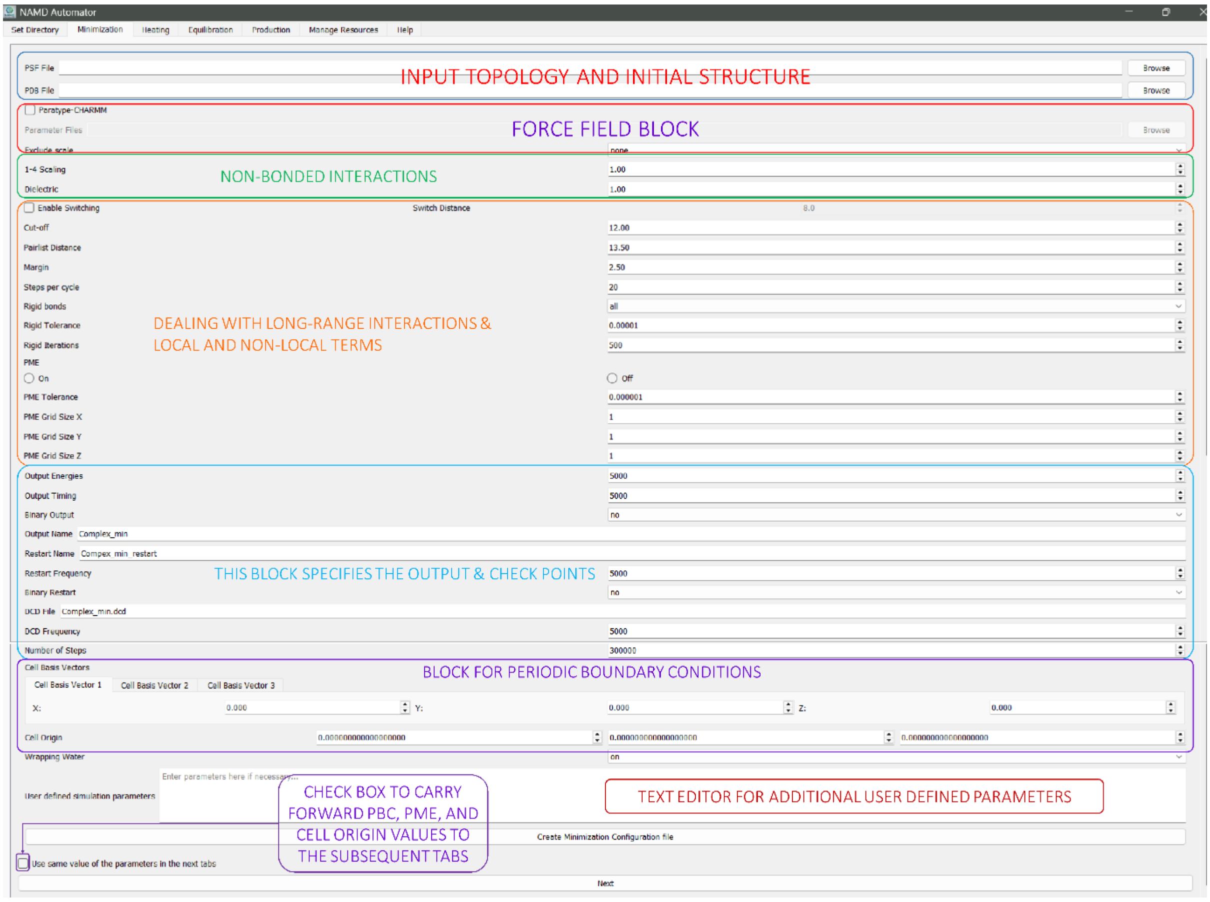
3. This action sets the working directory and its path is updated in the "Set Directory Tab". The following figure shows an updated "Set Directory Tab" with path of the working directory.



Once the working directory is established, the user can start working forward towards setting up the MD simulations. The first step in the CMD protocol is to setup the minimization simulation, the details to which are provided in the next section.

2. Minimization Tab

In CMD protocol, the very first step is to perform energy minimization. The objective to perform this stage is to ensure that there are no unfavorable contacts between the solvent molecules and the protein system under consideration. For performing this, and subsequent heating, equilibration, and production tasks in NAMD, a configuration file is essentially required. Such configuration files contains necessary Intel to conduct the simulation in NAMD. Minimization and subsequent tabs are designed to craft a suitable configuration file for each respective stage in CMD protocol. The default form of Minimization tab is shown as below:



<u>Features of the Minimization Tab</u>

The Minimization Tab is a single form that can be divided into six section.

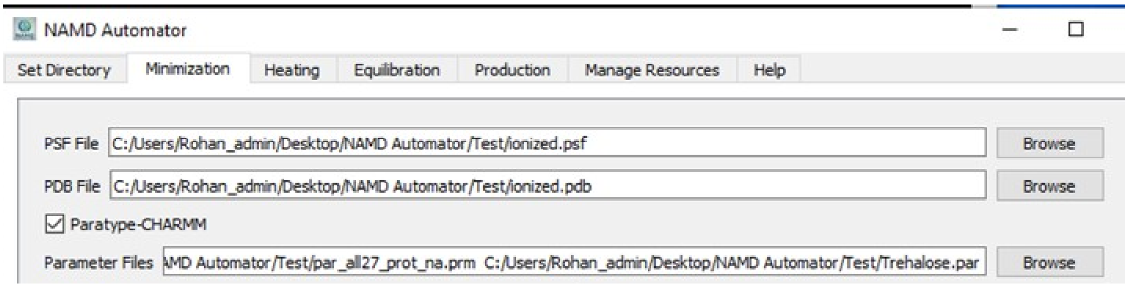
1. <u>Input Section:</u> The first section is designed to take input topology and coordinate files from the user through the PSF and PDB fields respectively. Browse buttons can be used to locate and select the respective files to load in the software
2. <u>Forcefield Block:</u> This section provides feature to load the force field files that are to be used in the simulation. The user is expected to upload all the relevant parameter files necessary for the simulation.

3. Non Bonded Interactions: This block offers the user the utility to define how non bonded interactions are to be dealt.
4. Long Range Interactions: This section enables the user to decide how long range interactions are defined and dealt during the simulation. It also provides features on defining local and non-local term values including cutoff values and switching distances, etc.
5. Output and check points: This section enables the user to define the frequency of saving the frames in the trajectory, name of the output files, and define the check points that will save the backup after regular intervals. It also specifies the frequency at which the energy data will be logged in the resulting Log file.
6. Periodic Boundary Conditions: The periodic boundary conditions can be defined and set using this block. The X, Y and Z coordinates of the three Cell Basis Vectors can be populated in the system through respective sub tabs (Cell Basis Vector 1, Cell Basis Vector 2, and Cell Basis Vector 3 respectively). The X, Y and Z coordinates of the Cell Origin can be specified in the “Cell Origin” field.
7. User Defined Parameters: Besides these necessary components to conduct the simulations, we provide a separate text box that can be used to include extra parameter in the simulation. For example, user can include information for fixing atoms, applying constrains or restrains etc.
8. The Carry Forward Utility: We offer a check box named “Use Same Value of the parameters in the next tabs”. This feature enables the user to carry forward the values of PBC, PME, Cell Origin, and other relevant values, and selections to the next and subsequent tabs.

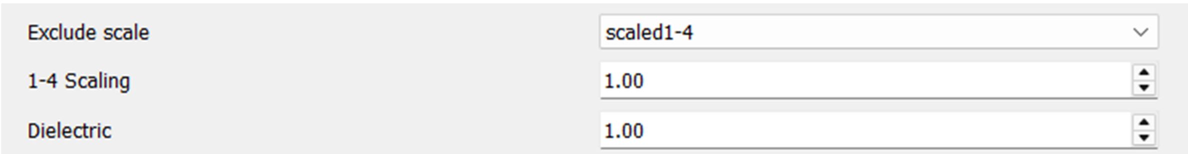
Utility of the Minimization Tab

After setting the working directory, user needs to upload the topology and the coordinate files into the software. The topology and coordinates files are uploaded in the form of PSF and PDB files respectively. This task of locating the respective files can be done via two Browse buttons in the “Input section” of the form.

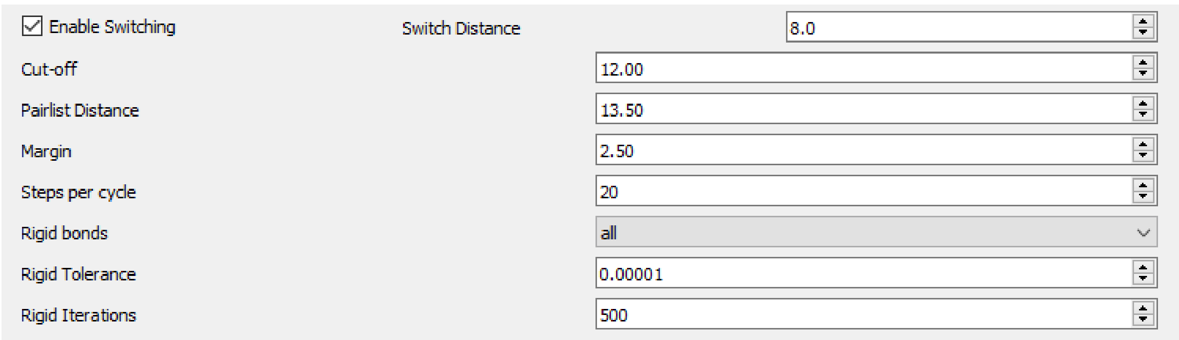
To successfully launch the simulation, appropriate parameter files are required in NAMD. Such parameter files can be included in the simulation using the “Forcefield Block”. The force field block needs to be activated by selecting the “Paratype-CHARMM” checkbox. Once activated, the field named “Parameter Files” becomes available along with the “Browse” button now functional. The user can now locate and upload the parameter files in the software. Multiple parameter files can be selected using the control button. Once all the input files are loaded the screen might look as follows:



The subsequent "Non Bonded interaction" section provides the features to select the "Exclude Scale" function in NAMD. In the current release of the software we are offering the 1-4 Scaling parameter to include in the simulation, as it is most preferred in the biological context. 1-4 Scaling decides the Scale factor for (i, i+3) electrostatic interactions. The users are advised to use the default values of 1 for "1-4 Scaling" and "Dielectric" fields as shown in figure below.

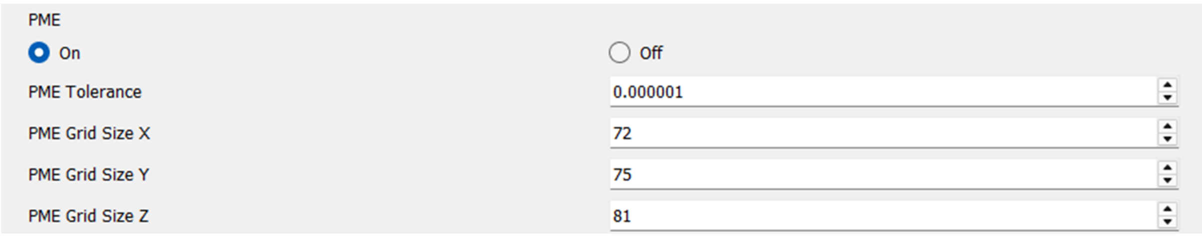


The next section dealing with the long range interaction sets the parameters for local and non-local terms. The same section sets up the bond constrain settings as shown in the figure below.



Ticking the "Enable Switching" check box will turn on the Switching Function in NAMD and set the value of 8 as default "Switch Distance". As per the value of Switching Distance, the default values of 12 and 13.5 are assigned to "cutoff" and "Pairlist Distance" parameter. A buffer margin value of 2.5 around the simulation cell is suitable for most of the solvated protein systems. Steps per cycle is set to the default value of 20. It signifies that after every 20 MD steps the full force evaluations will happen and neighbor list will be updated. The remaining three values in the section decides the bond constrain settings parameters. For instance, Setting “Rigid bonds” to value ‘all’ means that during the simulation, all bonds including the hydrogen atoms will be constrained. Rigid Tolerance value corresponds to the convergence criteria for the rigid bonds. Rigid Iteration value enables NAMD to evaluate the value for number of iteration for the constraint solver algorithm. Generally, SHAKE or RATTLE algorithms are used.

The subsequent section (see figure below) sets up the Particle Mesh Ewald (PME) parameters to be used in the simulation.



The software offers to use, PME method to handle long range electrostatic interactions in NAMD. For implementing the PME, a suitable grid is needed to be defined. The PME grid size is set by providing the values in the three dimension (X, Y and Z). PME Grid Size X, Y and Z will differ from system to system. The values of PME Grid Size X, Y and Z for a specific system under consideration can be obtained by running the PBC-PME calculator script. The PME Tolerance value corresponds to the accuracy of computing the Ewald real space.

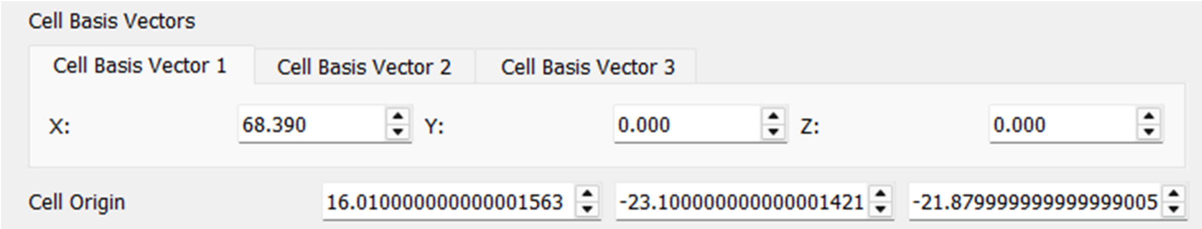
The following “Output and Check points” section (See figure below) instruct NAMD about how and when frames are to be stored in trajectory files? How and when backup files are to be generated? What should be the names of these files?

The first two fields in this section, “Output energies” and “Output timing” corresponds to the interval in the form of time steps. The “Output energies” is an interval at which all the energy terms are written in the output log file. The “Output timing” corresponds to the interval to report the timing in the log file that covers speed and memory allocation. The field “Binary output” defines if the last structure is saved in a binary file. The text content written in the “Output Name” field decides the names of the output files where velocities, coordinates and extended system information is saved. Three files with extension .vel, .coor and .xsc will be generated with the name specified in this field at the end of the simulation. **The users are advised to keep the names of the output files to default in order to maintain the uniformity of file names and thereby ensure the automation in the process.** Similarly, the restart files (checkpoints/backup files) for velocities, coordinates and extended system information with the user defined names are generated every time the “Restart Frequency” interval value is reached in the simulation. The files generated with “Restart Frequency” names are updated at the regular interval during the simulation. Hence, these files are of utmost importance if the user needs to extend the simulation or restart the simulation due to power failure. Decision made in “Binary Restart” field will reflect if the restart files are in binary form or not. The field “DCD file” provides the name to the resulting trajectory file in dcd format. The “DCD frequency” value decided the interval at which the frames are recorded in the trajectory.

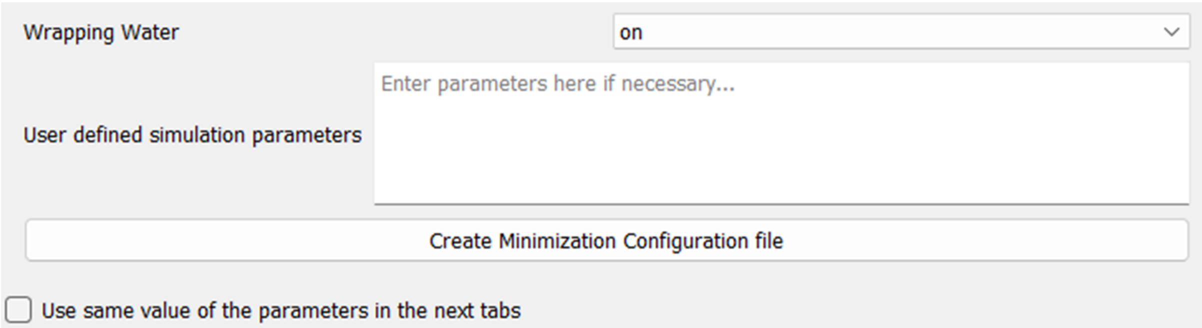
The “Number of Steps” field dictates the time duration of minimization runs. It corresponds to the number of conjugate gradient minimization steps.

The final section defines the PBC and Wrapping Function. To setup the PBC, three Cell Basis Vectors (CBV1 to 3) are needed to be defined. These three vector corresponds to the X, Y and Z directions for the unit cell. These values are also specific to every system and hence they are needed to be updated when the protein system changes. Values for three CBVs can be obtained by running the script PBC-PME calculator. The X, Y and Z values for each CBV can be entered via separate subtab named “Cell Basis Vector 1”, “Cell Basis Vector 2”, and “Cell Basis Vector 3”.

The “Cell Origin” field defines the coordinates to the position of the unit cell center. The three coordinates of cell origin can be obtained by running the script PBC-PME calculator. The decision made in “Wrapping water” dictates if the water molecules are translated back to the unit cell or not.



The “User Defined Simulation Parameters” field can be used to include additional parameters for running the simulation.



Once all the fields are populated in the form, pushing the “Create Minimization Configuration file” button will open the dialogue box for saving the configuration file. The user has a chance to name the file as per his/her requirement. Pushing the “Save” button creates a configuration file and saves in the working directory. By default, the minimization configuration file is saved by name “minimization.conf”.

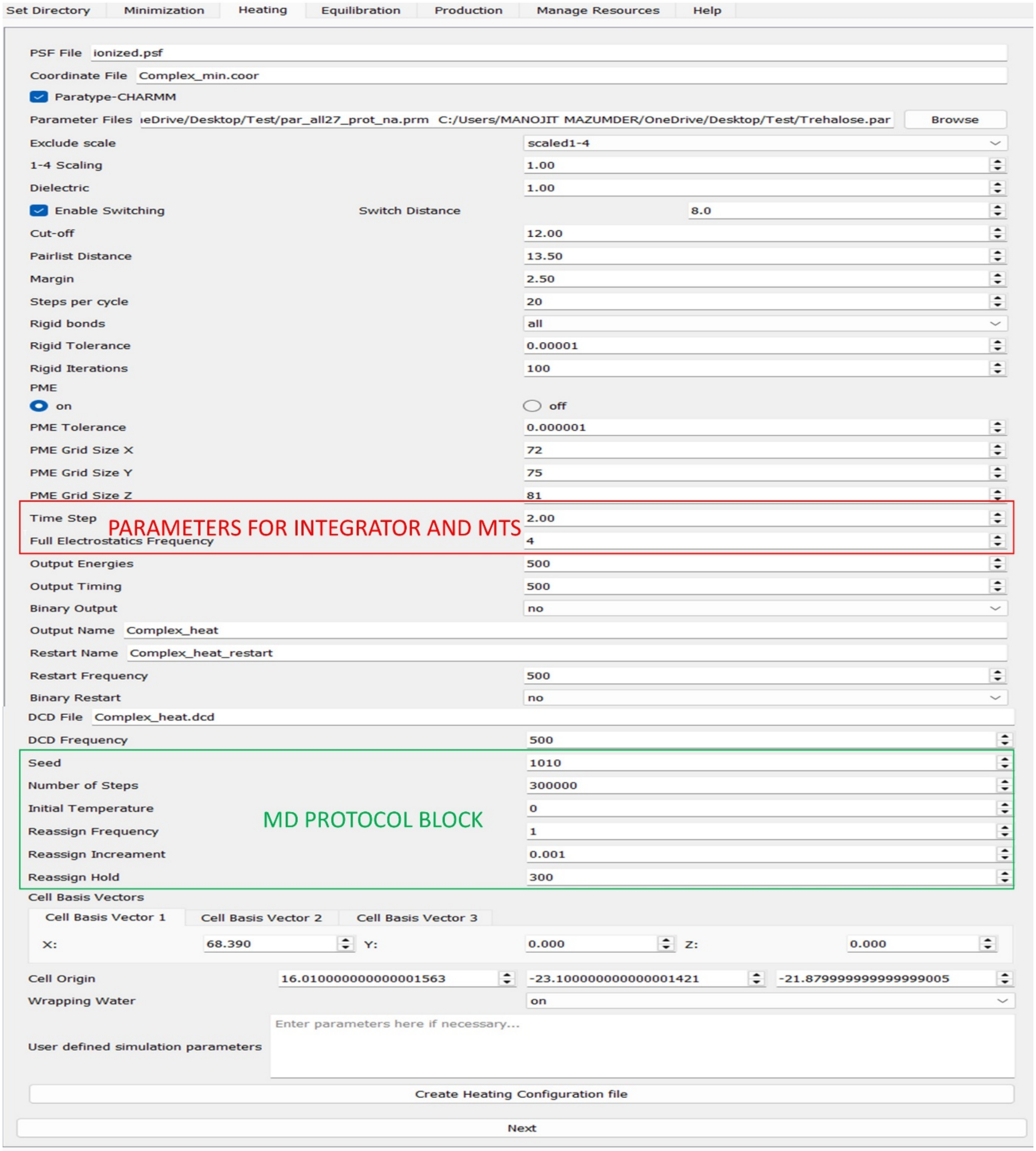
The sample configuration file generated is shown in figure below:

```
##############################################
#### input topology and initial structure ####
##############################################
structure        ionized.psf
coordinates      ionized.pdb

##############################################
#### force field block #######################
##############################################

paratypecharmm on
parameters        par_all27_prot_na.prm
parameters        Trehalose.par
exclude             scaled1-4

##############################################
#### non-bonded interactions #################
##############################################

1-4scaling     1.0
dielectric      1.0

##############################################
### dealing with long-range interactions######
##############################################

switching              on


##############################################
#### local and nonlocal terms   ##############
##############################################

switchdist           8.0
cutoff                12.0
pairlistdist         13.5
margin             2.5
stepspercycle      20
rigidBonds         all
rigidTolerance     0.00001
rigidIterations    500

############################################
#### local and nonlocal terms   ############
############################################

PME                on
PMETolerance       0.000001
PMEGridSizeX       72
PMEGridSizeY       75
PMEGridSizeZ       81
minimization         on

############################################
###### this block specifies the output #####
############################################

outputenergies     500
outputtiming       500
binaryoutput       no
outputname         Complex_min
restartname        Compex_min_restart
restartfreq        500
binaryrestart      no
DCDfile            Complex_min.dcd
dcdfreq            500
numsteps           6000

######################################################
# this block defines periodic boundary conditions ####
######################################################
cellBasisVector1     68.39 0.0 0.0
cellBasisVector2     0.0 74.25 0.0
cellBasisVector3     0.0 0.0 82.05
cellOrigin           16.01 -23.1 -21.88

wrapWater            on
```

2. Heating Tab

Heating corresponds to the process where the temperature of the biomolecular system is brought to nearly 300 K, which nearly corresponds to an average temperature where most of the protein systems are physiologically active. In this phase of CMD, the temperature of the system is gradually increase starting from 0 till 300 K. At each time step, the temperature would be increase by 0.001K. In order to gradually increase the temperature, various ‘Temperature reassignment parameters’ are needed to be specified in the configuration file. All these features are dealt in the Heating Tab (See figure below) of the NAMD-Automator software.

Features of the Heating Tab



The form for setting up the heating simulation consist of the following sections

1. Input Section: The first section is designed to specify the name of the input topology via the PSF field. However, the change is needed to be made in the coordinate file field. Since the heating procedure is to be started at the point where minimization was completed, the coordinate file saved at end of the minimization simulation is needed here. If the user sticks the default output file names in the previous minimization step, then the file named ‘Complex_min.coor’ corresponding to coordinate file written by NAMD at last timestep

of minimization simulation will be automatically generated during minimization and used in the current Heating stage.

2. The subsequent Forcefield Block, Non Bonded Interactions, Long Range Interactions, Output and check points section, Periodic Boundary Conditions block, and User Defined Parameters sections are the same as defined in the minimization tab. If the user has ticked in the check box named “Use Same Value of the parameters in the next tabs”, the corresponding values filled in the previous minimization tab might be automatically populated in the heating tab. This automation saves the time and efforts of manually filling the new form.
3. Parameters for integrator: In the heating tab, a new section that defines parameters for integration algorithm is introduced. The two new fields “Time Step” and “Full Electrostatic Frequency” are updated in this form.
4. Another new section named “MD Protocol Block” is updated at this stage. This section is the heart of the heating process. It consist of various fields like “Seed”, “Number of steps”, “Initial Temperature”, “Reassign Frequency”, “Reassign Increment”, and “Reassign Hold”

Utility of the Heating Tab

Except the change in the coordinate file name in the “data input block”, the data in the following blocks is carried forward. For example, values in the Forcefield Block, Non-Bonded Interactions, Long Range Interactions, Output and check points section, Periodic Boundary Conditions block, and User Defined Parameters sections are the same as defined in the minimization tab. Here we focus on utility of the new fields added in the Heating form.

As mentioned above, the Parameters for integrator section has two new fields, “Time Step” and “Full Electrostatic Frequency”. To maintain biological relevance, the time step value is set to 1 fs by default. The value for field “Full Electrostatic Frequency” is set to 4 fs. The value of 4 indicate that the full electrostatic evaluation will be conducted after every 4 fs.

In the output block, the corresponding changes have been made to name the output files that shall reflect heating procedure.

The new section introduced in this tab is the MD protocol block. This is heart of this configuration file and eventually the heating process. In the field named “Seed” the user need to specify one random number that will be used to assign initial velocities. The next field ‘Number of steps’ specifies the number of time steps for running the simulation. The heating simulation is needed to run for 300 pico seconds, therefore 300000 time steps will be required if time step of 1 femto seconds is considered. Therefore, total simulation time for any simulation can be calculated as

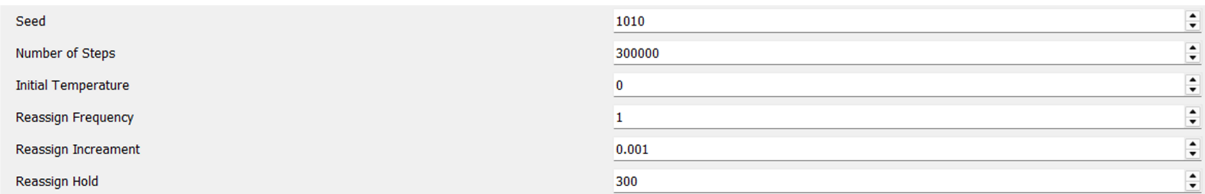
$$Total\ Simulation\ time = Numsteps\ \times timesteps$$

The field “Initial Temperature” indicate the initial temperature of the system at start of simulation. By default, initial temperature value is set to zero.

The user needs to specify that after how many timesteps the temperature is to be increased. Setting up the value in field “Reassign Frequency” to 1, means that temperature is to be increase after every one femto second.

NAMD uses simulated annealing protocol for increasing the temperature. The field “Reassign Increment” specifies how much temperature is to be increases at each timestep. The default value of 0.001 indicate that the temperature increment will take place by 0.001 K at each passing timestep. Therefore, in order to reach 300K, the heating simulation will require 300000 timesteps. Hence the default time for the heating simulation is set to 300 Ps.

The field “Reassign Hold” specifies NAMD program on where to stop increasing the temperature. Setting up value of 300 means no more incrementation in temperature once 300K is reached.



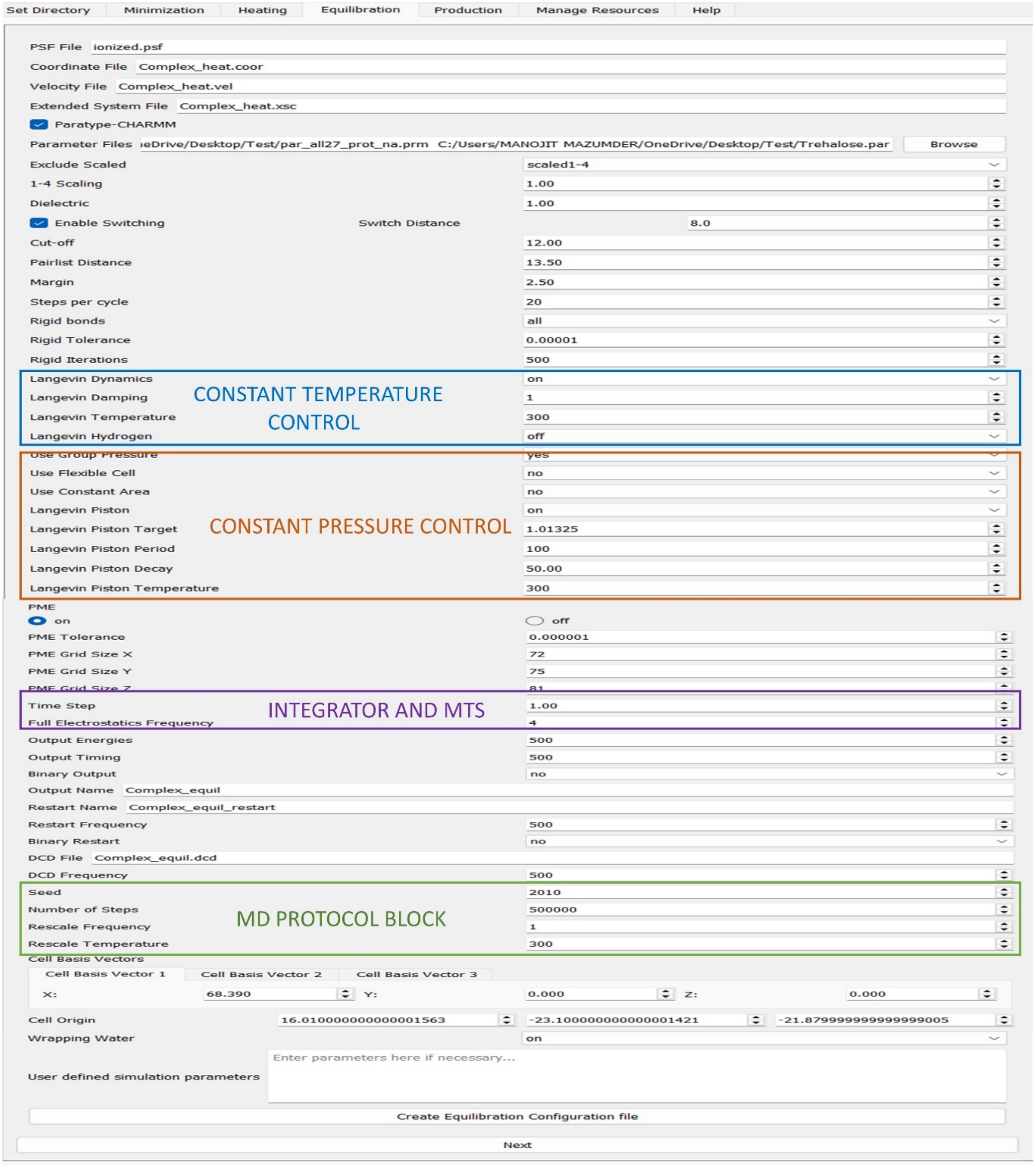
Since the same box is used for heating and further processes that we created for minimization, the values for PME and PBC will remain the same as in minimization Tab and hence configuration file.

After Filling up the relevant fields from the Heating form discussed above, pressing the "Create Heating Configuration file" button creates a configuration file and saves it in the working directory. By default, "heating.conf" file will be generated (see figure below).

```
#################################################
#### input topology and initial structure ####
#################################################
structure          ionized.psf
coordinates        Complex_min.coor


#################################################
#### force field block #######################
#################################################

paratypecharmm on
parameters         par_all27_prot_na.prm
parameters         Trehalose.par
exclude              scaled1-4
1-4scaling          1.0
dielectric           1.0


#################################################
### dealing with long-range interactions######
#################################################

switching                on
switchdist             8.0
cutoff                 12.0
pairlistdist           13.5
margin                 2.5
stepspercycle          20
rigidBonds             all
rigidTolerance         0.00001
rigidIterations        100

#################################################
### Ewald electrostatics######################
#################################################

PME                    on
PMETolerance           0.000001
PMEGridSizeX           72
PMEGridSizeY           75
PMEGridSizeZ           81


#################################################
### parameters for integrator and MTS ########
#################################################

timestep                   1.0
fullElectfrequency         4


#################################################
### the output ##############################
#################################################

outputenergies        500
outputtiming          500
binaryoutput          no
outputname            Complex_heat
restartname           Complex_heat_restart
restartfreq           500
binaryrestart         no
DCDfile               Complex_heat.dcd
dcdfreq               500


#################################################
### MD protocol block ########
#################################################

seed                    1010
numsteps                300000


temperature             0
reassignFreq            1
reassignIncr            0.001
reassignHold            300


##########################################################
# this block defines periodic boundary conditions #######
##########################################################
cellBasisVector1        68.39 0.0 0.0
cellBasisVector2        0.0 74.25 0.0
cellBasisVector3        0.0 0.0 82.05
cellOrigin              16.01 -23.1 -21.88

wrapWater               on
```

3. Equilibration Tab

The final goal of performing molecular dynamics simulation is to sample the structural states of the biomolecular system at equilibrium. That is what is intended to be done in the final production (data collection) phase. During previous steps like solvation, addition of ions to neutralize the charges and subsequent minimization and heating, the protein solvent system cannot be definitely assumed to be in equilibrium. All these processes add artifacts during system construction process that are necessary to be removed before the data is collected for analysis. In short, the configuration of system at this stage cannot be trusted to initiate data collection. For example, when the temperature of system is increased to 300K, the kinetic energy is introduced to the system thereby driving system away from equilibrium state. Therefore, during equilibration phase, the intention is to bring kinetic as well as potential energy at same level before data collection begins. Equilibration is generally carried out in NPT ensemble state. All these tasks are to be done using the Equilibration Tab (see figure below).



<u>Features of the Equilibration Tab</u>

Conceptually, there are 9 sections in the form. At this stage, emphasis will be given to the section where changes are to be made. The remaining sections will be assumed to have the same features and utility as discussed in the previous two Tabs.

In the equilibration, tab the input section is highly elaborated and deserves the attention. Like heating tab, the coordinate file generated at the end of previous heating phase has been used

(Complex_heat.coor). Additionally, two more fields are added here with names "Velocity File" and "Extended system file".

In the CMD protocol implemented in NAMD-Automator, the equilibration is to be conducted in the NPT ensemble, where temperature and pressure are to be maintained. For maintaining constant pressure and temperature in the equilibration, two more sections are added in the equilibration form.

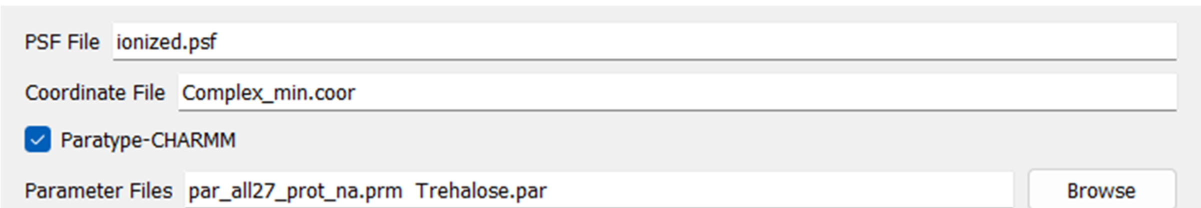
For maintaining constant temperature, NAMD uses Langevin dynamics method; and hence the same is used here. Block of fields with names "Langevin Dynamics", "Langevin Damping", "Langevin Temperature", and "Langevin Hydrogen" are implemented in constant temperature control.

NAMD uses Langevin method for maintaining constant pressure. This method uses Langevin dynamics for controlling fluctuation in barostat. Fields with names "Use Group Pressure", "Use Flexible Cell", "Use Constant Area", "Langevin Piston", "Langevin Piston Target", "Langevin Piston Period", "Langevin Piston Decay", and "Langevin Piston Temperature" are used for efficient pressure control.

The final changes are in the MD protocol block, where two more fields are added with names "Rescale Frequency" and "Rescale Temperature" that performs the actual equilibration of the system.

<u>Utility of the Equilibration Tab</u>

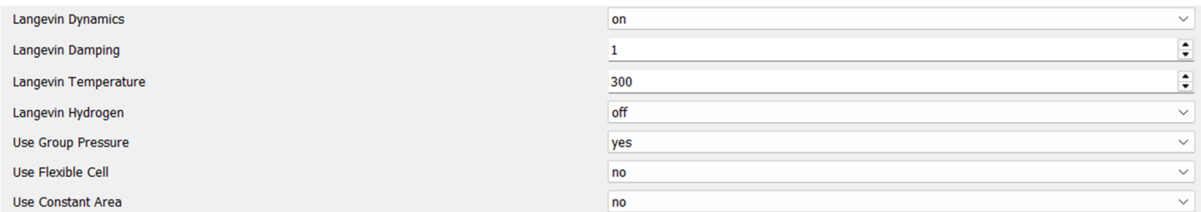
In the input section of the equilibration, the equilibration stage is needed to start right at the point where the heating has finished. In order to maintain the continuity, the coordinates, velocities, temperature, and periodic box information at that exact stage is required. Velocity file generated at end of heating stage stores data on particle velocities as well as temperature, while xsc file holds data on periodic boundary condition. Hence these files are needed to start the equilibration phase.



No extra features are introduced in the blocks for forcefield, Ewald electrostatic, output, long range interaction and PBC. However, in the output block, changes have been made to the names of the output files that reflect equilibration procedure.

In the temperature control block, the decision made in the field “Langevin Dynamics”, decides to enable or disable the Langevin Dynamics protocol in NAMD. In short, the Langevin Dynamics protocol implemented in NAMD simulates the effect of heat bath. Such an effect is artificially generated by adding random and frictional forces to the atoms, which are sometimes referred to as Stochastic and Damping forces respectively. In the field “Langevin Damping”, the default value 1 sets the Damping coefficient Gamma to 1 ps^{-1}. The value less than 1 might reflect weaker coupling to the heat bath, while values greater than 5 might mean stronger coupling. The default value 1 is the most optimal value for maintaining the temperature of biological interest. The field “Langevin Temperature” is meant to set the target temperature of the simulated thermostat. The default value of 300 in this field means that the biomolecular system will be regulated around 300 K during the equilibration process.

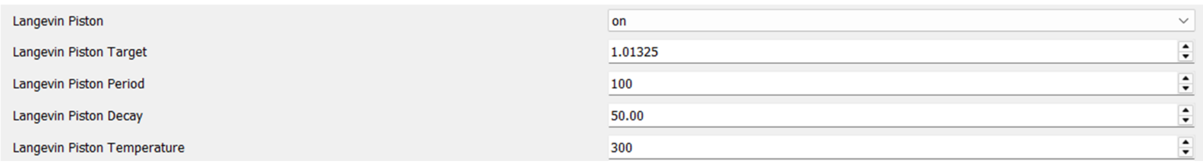
In NAMD-Automator, the Langevin Piston pressure control method is implemented. The first field “Use Group Pressure” if enabled instructs NAMD to use the group-based contribution for pressure calculations rather than single atoms. It is mandatory to use group pressure utility when “Rigid bonds” are activated for all atoms including hydrogens in the Constraints settings. Using the value “No” in the field “Use Flexible Cell” means that the shape of the simulation box will not change. It will be assumed that the pressure calculations will be isotropic, i.e the box can either expand or contract equally in all direction. Isotropic pressure calculation are recommended when the system is Isotropic. The simulation of the globular protein in a solution is a typical example of isotropic system. In short, the CMD protocol described in NAMD Automator best suits the biomolecular systems where globular proteins are solvated. Initiating the value “No” in the “Use Constant Area” is related to the choice made in the field “Use Flexible Cell”. Setting it to no would mean that the area can change. Using Yes value in “Use Constant Area” would be more appropriate in Membrane based simulations.



The remaining fields in the block pertain to the “Langevin Barostat” that maintain constant pressure (Similar to “Langevin Thermostat” that control temperature). Here, NAMD simulates a piston variable that couples to volume change and thus control the pressure in the system.

In the field “Langevin Piston”, the default value “On” indicate enabling the usage of Langevin Piston method to simulate the artificial barostat. The default value of 1.01325 in the field “Langevin Piston Target” is the target pressure value in Bar (equivalent to 1 Atm). During the simulation, the system volume is set to adjust dynamically to maintain the pressure of 1 Atm.

Setting the default value of 100 in the field “Langevin Piston Period” means that the Oscillation period of 100 fs for the simulated piston. This value decides how fast the system will respond to the fluctuation in pressure in the simulation. 100 is a moderate value suitable for most of the solvated globular biomolecular systems. The field “Langevin Piston Decay” expects the value of the Damping Coefficient (alternatively referred to as friction coefficient) in fs time scale. This value controls the decay time of the piston oscillation. The final field in this block, “Langevin Piston Temperature” connects the thermal bath temperature in the thermostat setting. This value must match the simulation temperature, in the default case, 300 K.

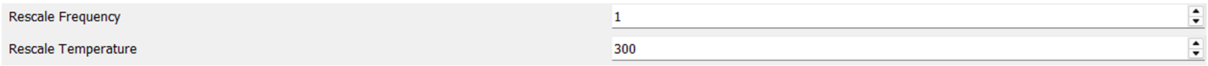


The take home message is that the default values in temperature and pressure control are well calibrated for a typical biomolecular system. If the user decides to change the setting, then corresponding lines are needed to be updated using the blank “User defined simulation parameters” box.

In final MD protocol block two more features are introduced with field names “Rescale Frequency”, and “Rescale Temperature”. These two fields are meant to control the ‘Temperature rescaling parameters. These are the two fields in the Tab (and eventually configuration file) where actual pressure and temperature is maintained and system is equilibrated. NAMD equilibrates the system by temperature rescaling method.

The default value 1 in the “Rescale Frequency” field activates the equilibration function of NAMD. The value for this parameter decides the time interval between time steps for temperature rescaling.

The Value in the field "Rescale Temperature" expects the temperature at which rescaling will be done at every "Rescale Frequency" time step.

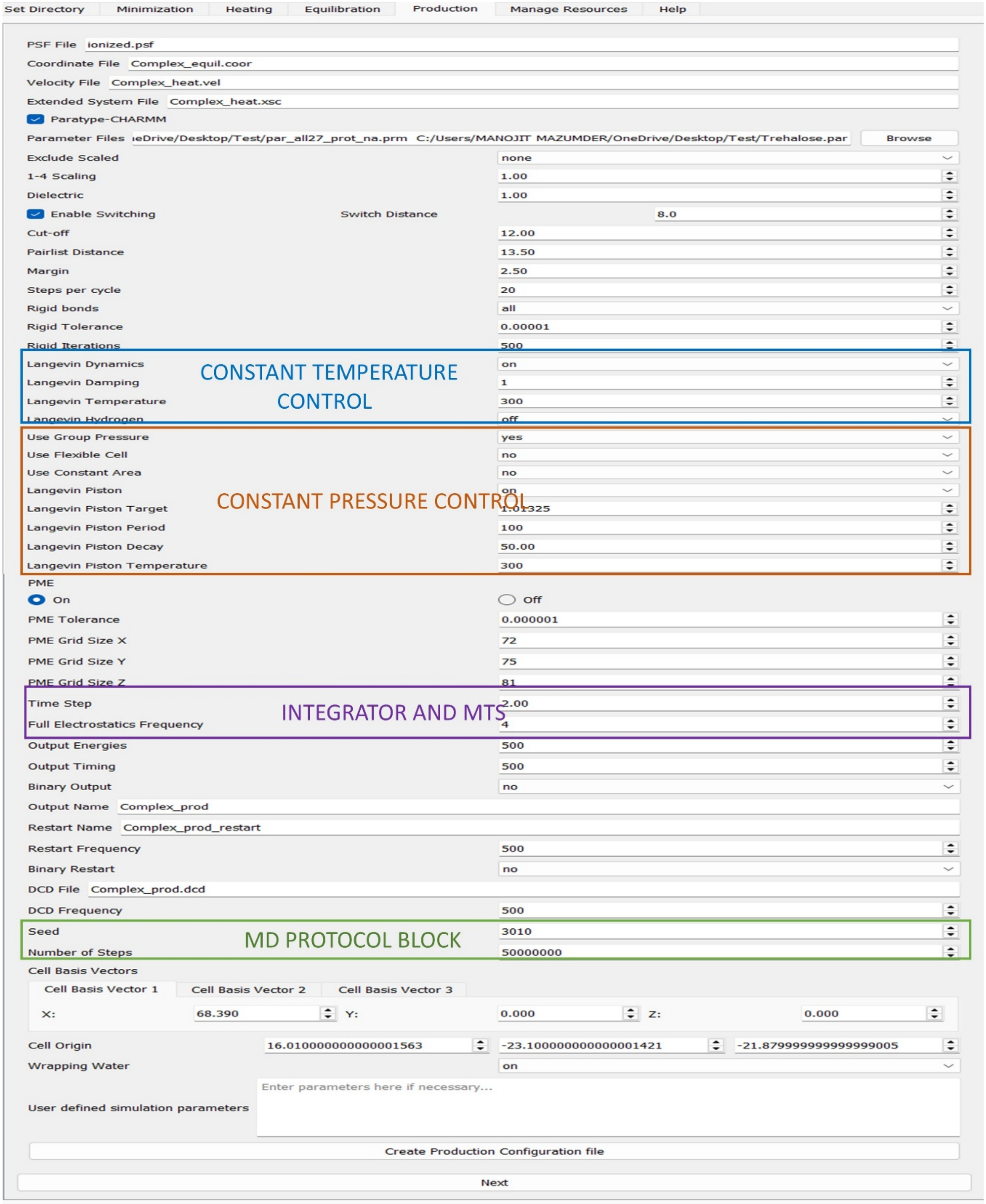


Pressing the "Create Equilibration Configuration File" button will compile all user provided data from the Equilibration tab into a configuration file and save it in the working directory.

```
#################################################
#### input topology and initial structure ####
#################################################
structure              ionized.psf
coordinates            Complex_heat.coor
velocities             Complex_heat.vel
extendedsystem         Complex_heat.xsc

#################################################
#### force field block ######################
#################################################

paratypecharmm         on
parameters         par_all27_prot_na.prm
parameters         Trehalose.par
exclude                scaled1-4

1-4scaling           1.0
dielectric            1.0

#################################################
### dealing with long-range interactions######
#################################################

switching              on
switchdist             8.0
cutoff                 12.0
pairlistdist           13.5
margin                 2.5
stepspercycle          20
rigidBonds             all
rigidTolerance         0.000010
rigidIterations        500

#################################################
###### Constant Temperature Control ##########
#################################################

langevin               on
langevinDamping        1
langevinTemp           300
langevinHydrogen       off

#################################################
###### Constant Pressure Control #############
#################################################

useGroupPressure            yes
useFlexibleCell             no
useConstantArea             no

langevinPiston               on
langevinPistonTarget         1.013250
langevinPistonPeriod         100
langevinPistonDecay          50.0
langevinPistonTemp           300

#################################################
###### Ewald electrostatics ##################
#################################################

PME                 on
PMETolerance        0.000001
PMEGridSizeX        72
PMEGridSizeY        75
PMEGridSizeZ        81

#################################################
###### parameters for integrator and MTS #####
#################################################

timestep                    1.0
fullElectFrequency          4

#################################################
###### the output ############################
#################################################

outputenergies              500
outputtiming                500
binaryoutput                no
outputname                  Complex_equil
restartname                 Complex_equil_restart
restartfreq                 500
binaryrestart               no
DCDfile                     Complex_equil.dcd
dcdfreq                     500

#################################################
###### MD protocol block #####################
#################################################

seed                2010
numsteps            500000
rescaleFreq         1
rescaleTemp         300

#######################################################
# this block defines periodic boundary conditions #######
#######################################################
cellBasisVector1          68.39 0.0 0.0
cellBasisVector2          0.0 74.25 0.0
cellBasisVector3          0.0 0.0 82.05
cellOrigin                16.01 -23.1 -21.88

wrapWater                 on
```

4. Production Tab

After completing the equilibration phase, the system is ready for the final phase of MD protocol where data is collected by sampling the structural characteristics and dynamics of the biomolecular system. As data from this phase is generally used for further analysis; therefore, it is called as production phase. The system setup for the production phase is conducted using the Production Tab in NAMD-Automator as shown in the figure below.



<u>Features of the Production Tab</u>

In NAMD-Automator, the production phase in NPT ensemble is offered. Therefore, all the features from the Production Tab are essentially required in the production. Fundamentally, removing the "Temperature rescaling parameters" that performs equilibration will be sufficient for initiating a production simulation. Therefore, in the production tab, the "Rescale

Frequency” and “Rescale Temperature” fields are absent that performs the equilibration and enable NAMD to run the production simulation in the NPT ensemble.

Utility of the Production Tab

Since the production run is supposed to start from fully equilibrated system, the coordinates, velocity and extended system information from coor, vel and xsc files respectively is required. These will be generated at the end of equilibration stage and will be present in the working directory by the time production phase start.

Therefore, the fields in the input section are already populated with the default names of the files required to initiate the production run.

Similarly, the corresponding changes have been made in the output block fields that names the output files and reflects the production procedure.

Finally, hitting the “Create Production Configuration file” button compiles the data from the Production Tab into a separate configuration file and saves the configuration file in the working directory.

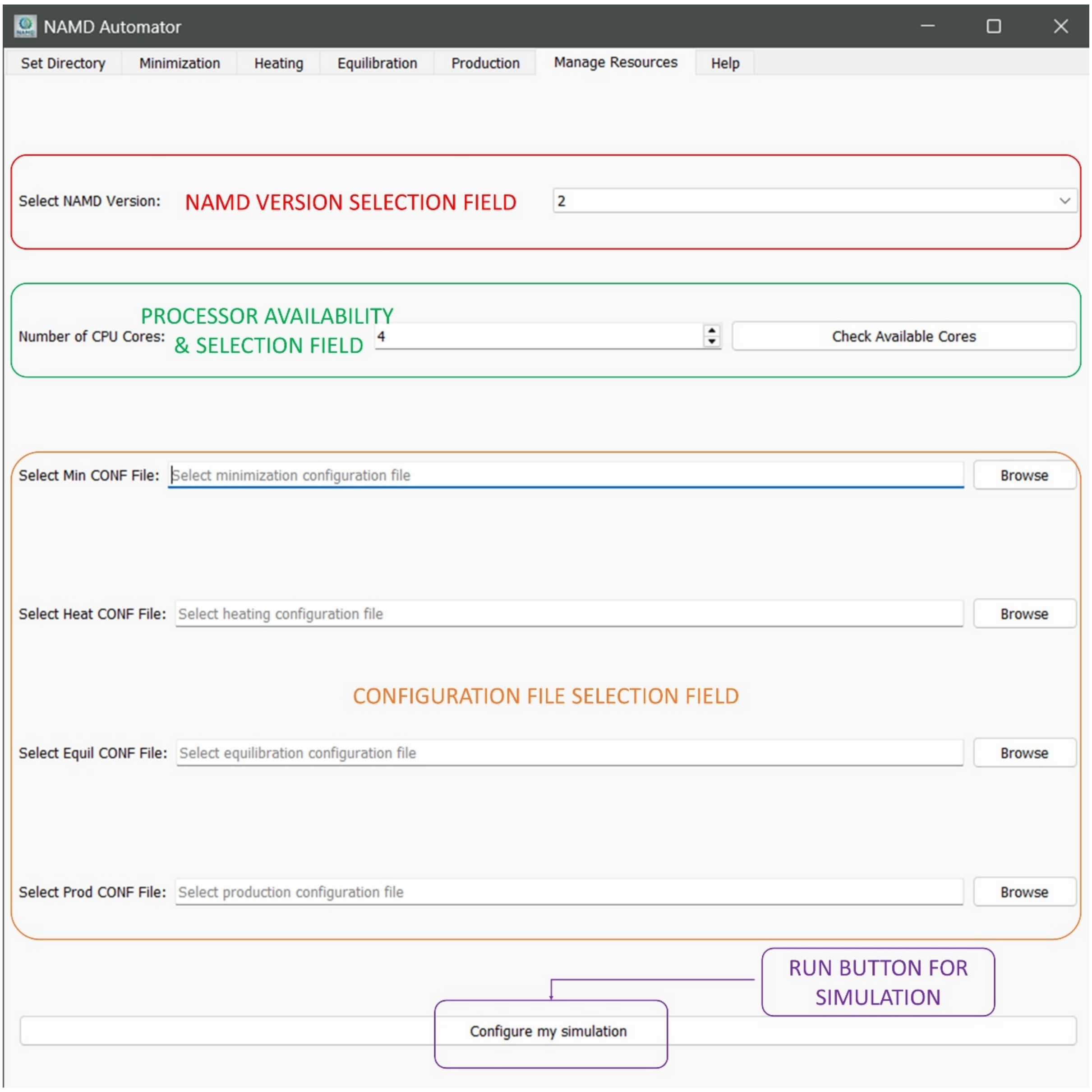
```
################################################
#### input topology and initial structure ####
################################################
structure           ionized.psf
coordinates         Complex_equil.coor
velocities          Complex_equil.vel
extendedsystem      Complex_equil.xsc


################################################
#### force field block ####
################################################
paratypecharmm          on
parameters          Trehalose.par
parameters          Trehalose.par
exclude scaled          scaled1-4


1-4scaling            1.0
dielectric             1.0
################################################
#### dealing with long-range interactions ####
################################################
switching                 on


switchdist                8.0
cutoff                    12.0
pairlistdist              13.5
margin                    2.5
stepspercycle             20
rigidBonds                all
rigidTolerance            0.00001
rigidIterations           500

################################################
#### Constant Temperature Control ############
################################################
langevin                  on
langevinDamping           1
langevinTemp              300
langevinHydrogen          off

################################################
#### Constant Pressure Control ###############
################################################
useGroupPressure               yes
useFlexibleCell                no
useConstantArea                no
langevinPiston                 on
langevinPistonTarget           1.013250
langevinPistonPeriod           100
langevinPistonDecay            50.0
langevinPistonTemp             300
################################################
#### Ewald electrostatics ####################
################################################
PME                 on
PMETolerance        0.000001
PMEGridSizeX        72
PMEGridSizeY        75
PMEGridSizeZ        81


################################################
#### parameters for integrator and MTS #######
################################################
timestep                      2.0
fullElectFrequency            4

################################################
#### the output ##############################
################################################
outputenergies                500
outputtiming                  500
binaryoutput                  no
outputname                    Complex_prod
restartname                   Complex_prod_restart
restartfreq                   500
binaryrestart                 no
DCDfile                       Complex_prod.dcd
dcdfreq                       500


################################################
#### MD protocol block #######################
################################################
seed                3010
numsteps            50000000

################################################
#### periodic boundary conditions ############
################################################
cellBasisVector1              68.39 0.0 0.0
cellBasisVector2              0.0 74.25 0.0
cellBasisVector3              0.0 0.0 82.05
cellOrigin                    16.01 -23.1 -21.88

wrapWater                     on
```

5. Manage Resources Tab

One of the most important objective to develop this software was to automate the MD simulation procedure, and offer utilities for efficient resource management to maximize the simulation performance. Keeping that in mind, we developed this tab with following objectives:

1. Identification of the available computational resources available on the machine and report it.
2. Create an efficient queuing system that enables sequential execution of the Minimization, Heating, Equilibration, and Production phases in an automated workflow without human intervention.

All these tasks are carried out in the "Manage Resources Tab" as shown in the figure below:



Features of the Manage Resource Tab

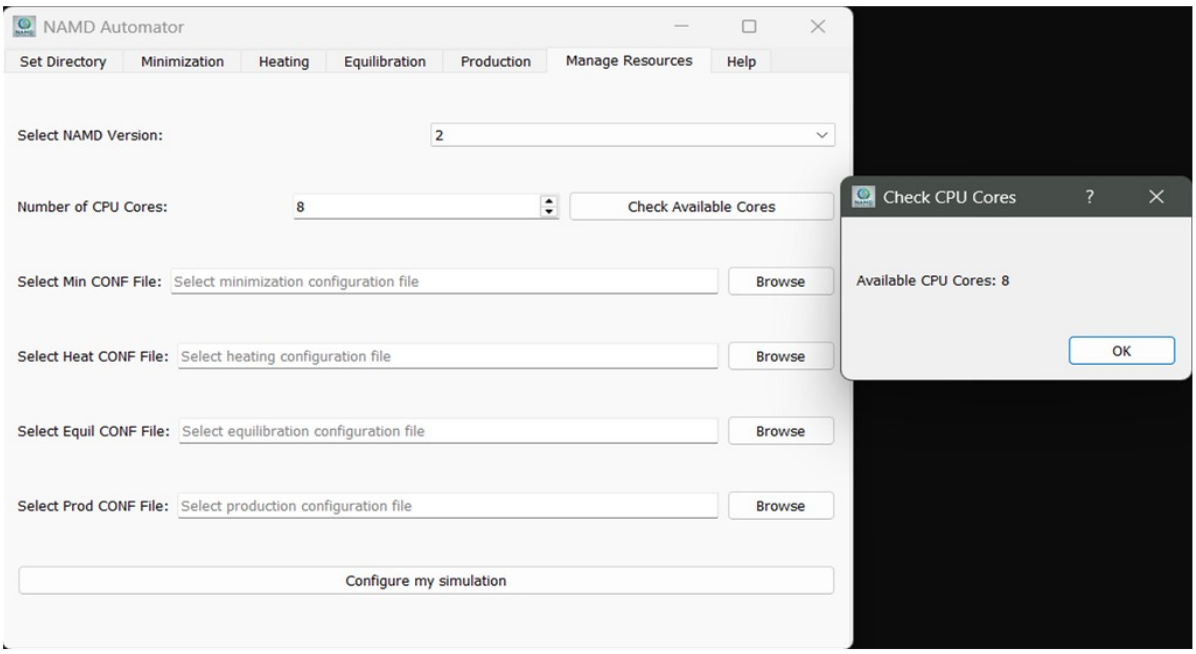
The first field, “Select NAMD Version”, enables the user to define the NAMD version to be used in the trajectory calculations. The second field “Number of CPU Cores”, is developed for identification of number of processors available for calculation and report them.

The subsequent four fields, “Select Min CONF File”, “Select Heat CONF File”, “Select Equil CONF File”, and “Select Prod CONF File” are intended to locate and upload the respective configuration files generated via previous four tabs.

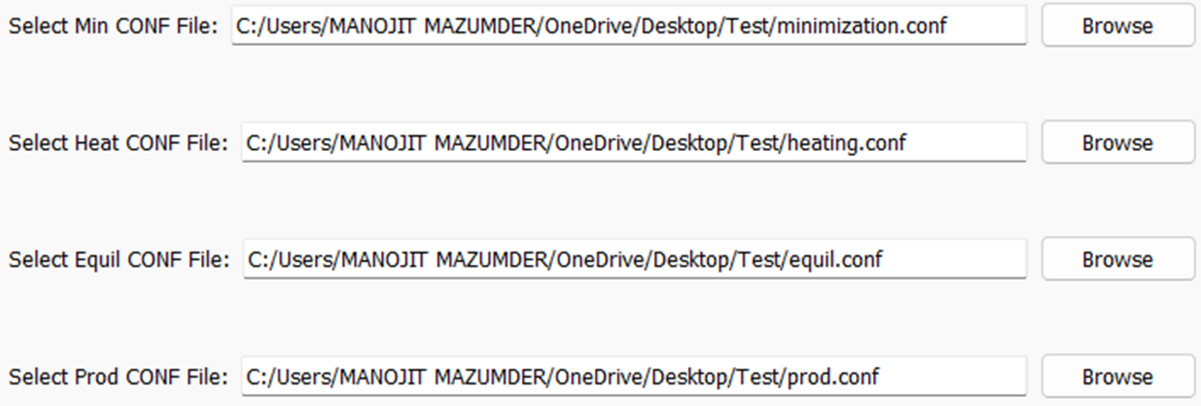
Utility of the Manage Resource Tab

By the time of writing this manual, NAMD version 2 and 3 were available for public use. Therefore, user can select the desired version that is installed on the machine through the dropdown menu and proceed for calculation.

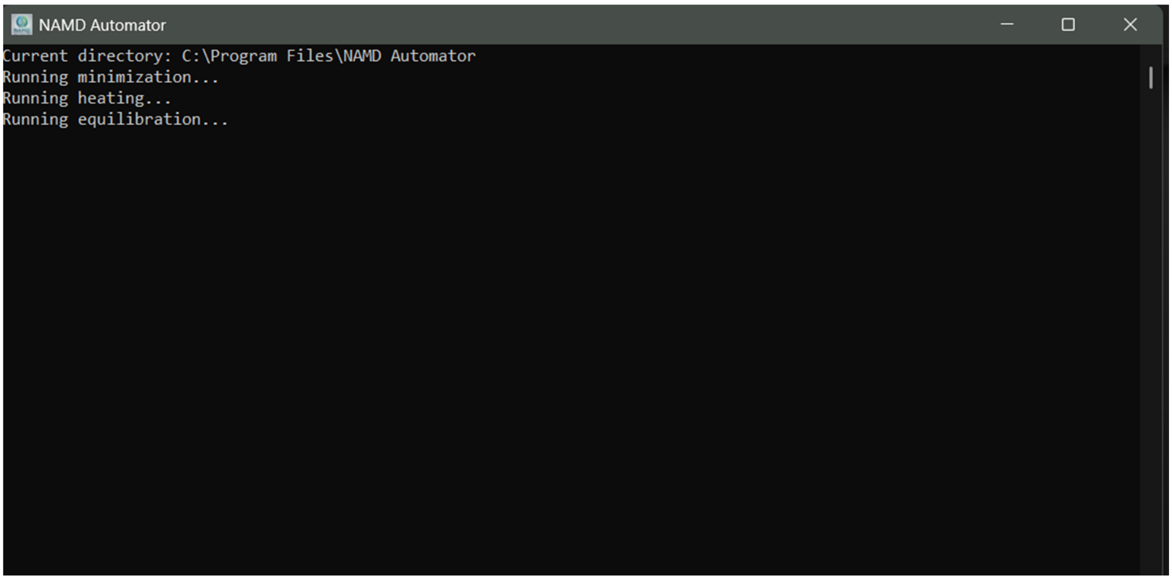
Pressing the “Check Available Cores” button in the “Number of CPU Cores” field opens a dialogue box reporting the number of available CPU cores in the machine. Pressing the OK button closes this box. Based on the number of available cores, the user can define the computational resources to allocate for MD simulations as per the requirement.



Finally, the user is required to load the appropriate configuration files generated for Minimization, Heating, Equilibration, and Production run through the “Browse” button in the respective fields.



Once, all the required configuration files are loaded in the software, pressing "Configure my Simulation" button creates a pipeline in the background and queues the simulations to logically execute one after the other. Once the pipeline is launched, the progress can be monitored in the command prompt as shown in the figure below:



The command prompt can be used to monitor the progress or perform trouble shooting for the error occurrences if any. The resulting log files will be generated in the working directory that can be used for subsequent trouble shooting.

The user has mobility to directly use the Manage Resource tab without generating the configuration files through the software, and launch the simulation work flow with user defined configuration files. This approach is applicable, provided the user has all the required configuration files crafted properly, and all the input files like final solvated PDB-PSF pair of the system, all parameter files are located in the working directory. While developing this system, we have made sure to segregate the functionality of GUI from the background pipeline that launch and conduct the simulations. As a result, multiple simulations can be setup and executed in parallel. However, this approach is recommended for the system with sufficient number of CPU cores, with good performance available for calculations. The GUI can be closed once the simulation pipeline is successfully launched in the command prompt. The command prompt needs to remain active until the simulations are finished.

Understanding files generated during simulation

Let us have a look at what files have been generated during the simulation. A lot of files appear to be populated in the working directory. Let us first understand what files are generated and what information is contained in them.

First and most important file is the .dcd file (Complex_min.dcd, and similar files for heating, equilibration, and production runs). These files contain all dynamics data, they are generally referred to as the trajectory file. It is binary file and contains frames of system saved at specified interval. Now, note that the dcd frequency value was set to 1000 in the minimization configuration file. Therefore, NAMD has saved structural state of the system after every 1000 steps. The minimization simulation was run for 6000 time steps, therefore there shall be six frames in the trajectory file. Since dumping all the structural information every time into dcd file in PDB format will increase the size of DCD file; therefore, the trajectory data is saved in the binary form that VMD program can understand.

Next in line, there are files with extension .coor (Complex_min.coor), .vel (Complex_min.vel) and .xsc (Complex_min.xsc). These files are generated as per the specified output name to start with "Complex_min" in the configuration file. Therefore, the same name has been appended to these files. File Complex_min.coor contains the coordinates of all atoms at the end of simulation.

File with vel extension is called as velocity file (Complex_min.vel). This file will contain velocities of every atom in the system at the end of simulation. The same file also saves temperature information of system.

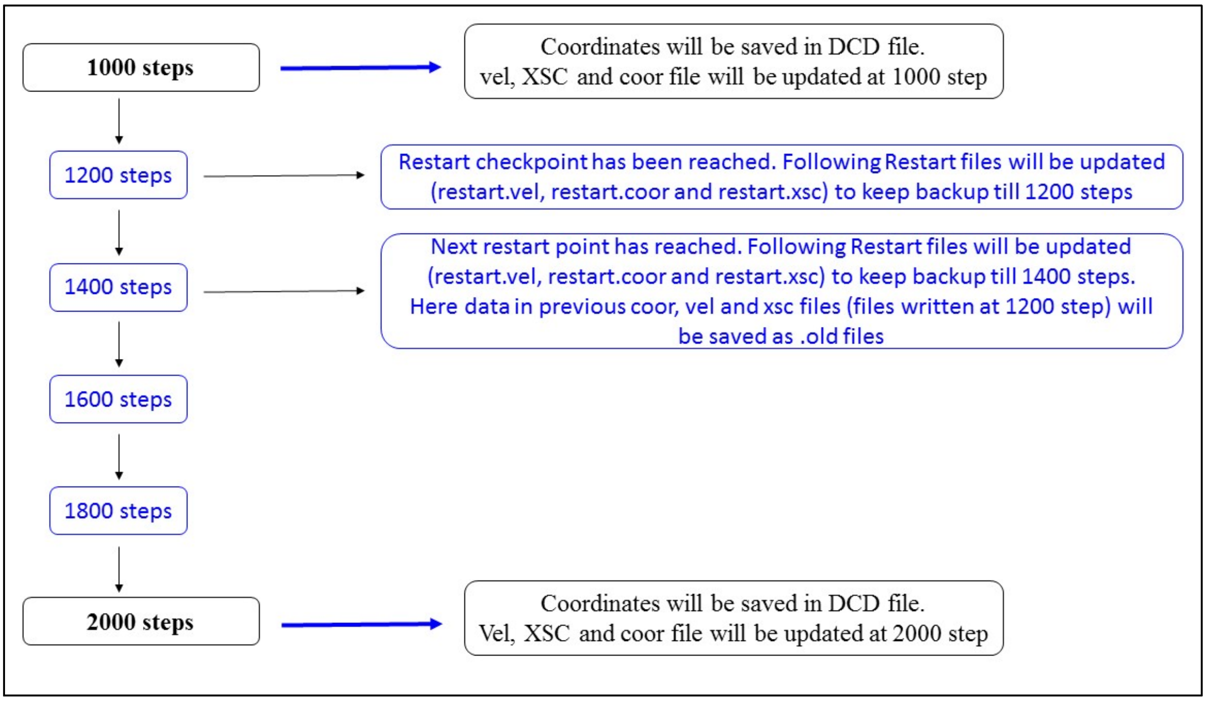
File with xsc extension is called as Extended System Configuration file (Complex_min.xsc). This file holds information of the periodic box and other system variable information. During execution of the constant pressure simulation (for example equilibration phase), the strain rates of the system will be saved in this xsc file. In short, most of the remaining parameters for running subsequent simulations are contained in this file.

Why NAMD generate these files?

The remaining files contains keyword 'restart'. For example, "Complex_restart_min.coor", "Complex_restart_min.vel" and "Complex_restart_min.xsc". NAMD provides an opportunity to save the backup of velocities, coordinates and other system information to be saved in restart files before next dcd frequency value is reached.

NAMD will update the DCD file after every 1000 steps. The backup files will be overwritten or updated every time the restart checkpoint in reached. Once the restart checkpoint is reached, the data saved in previous restart will not be immediately overwritten but saved in another file with old extension. Thus, restart files and old files will be updated at every restart frequency checkpoint is reached

Following figure illustrates the process if the dcd frequency is set to 1000 and restart frequency to 200.



Why NAMD does this?

Data from these files (restart or old files) are used to restart the simulation in case if the simulation is abruptly stopped or aborted due to power failures or any unforeseen events.

Additionally, as commanded while running the simulation, a log file is generated that keeps log of everything that happened during simulation. If there occur any problem or error while running the simulation, this file would be the one you would first look into.